

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 24/01/2026**
Time: 3 Hrs **GTM - 24** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Assertion (A): When lightning strikes an aircraft, the current flows over the metal surface and does not pass through the inside.

   Reason (R):- Charges reside on the surface of conductors resulting in zero electric field inside.

   A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

   B) (A) is correct but (R) is not correct

   C) Both (A) and (R) are correct but (R) is not correct explanation of (A)

   D) (A) is not correct but (R) is correct

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2. Assertion(A): Radius of nucleus (R) and its mass number (A) are related as

$$\sqrt{A}\,\alpha\,R^{\frac{1}{6}}$$

Reason (R) : Density of a nucleus is independent of its mass number.

A) (A) is correct but (R) is not correct
B) (A) is not correct but (R) is correct
C) Both (A) and (R) are correct and (R) is the correct explanation of (A)
D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

3. The unit of $\sqrt{\frac{2I\mu_0}{c}}$ (I - intensity of an electromagnetic wave, c-speed of light)

A) Vm B) NC C) tesla D) NC^{-1}

4. The dimension of $\sqrt{\frac{\varepsilon_0}{\mu_0}}$ is equal to that of

(μ_0 = vacuum permeability and ε_0 = vacuum permittivity)

A) Capacitance B) Inductance C) Resistance D) conductance

5. A small metal plate of work function W is kept at a distance 'd' from a singly ionized fixed ion, A monochromatic light beam is incident on the metal plate and photo electrons are emitted. Find the maximum wavelength of the light beam so that some of the photoelectrons may go round the ion along a circle.

A) $\lambda = \frac{8\pi\varepsilon_0 dW}{e^2 + 8\pi\varepsilon_0 dhc}$ B) $\lambda = \frac{e^2 + 8\pi\varepsilon_0 dhc}{8\pi\varepsilon_0 dW}$

C) $\lambda = \frac{8\pi\varepsilon_0 dhc}{e^2 + 8\pi\varepsilon_0 dW}$ D) $\lambda = \frac{e^2 + 8\pi\varepsilon_0 dW}{8\pi\varepsilon_0 dhc}$

6. **Assertion (A):** Areal speed of the planet is always constant.

Reason (R): If a planet of mass 'M' has angular momentum 'L' about sun, then its areal speed is given by $\frac{\mathbf{L}}{\mathbf{2M}}$

A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C) (A) is correct but (R) is not correct

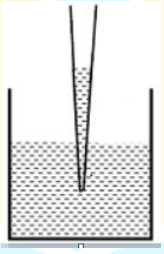
D) (A) is not correct but (R) is correct

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7. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T. Neglecting all vibrational modes, the total energy of the system is

A) 15RT B) 11RT C) 9RT D) 4RT

8. A glass capillary tube is of the shape of a truncated cone with an apex angle ‘α’ so that its two ends have cross-sections of different radii. when dipped in water vertically water rises in it to a height 'h', where the radius of its cross section is 'b'. If surface tension of water is 's' its density is ρ, its contact angle with glass 'θ' the value of ‘h’ will be [‘g’ is the acceleration due to gravity]



A) $\frac{2S}{b\rho g}\cos(\theta-\alpha)$ B) $\frac{2S}{b\rho g}\cos(\theta+\alpha)$

C) $\frac{2S}{b\rho g}\cos\left(\theta-\frac{\alpha}{2}\right)$ D) $\frac{2S}{b\rho g}\cos\left(\theta+\frac{\alpha}{2}\right)$

9. An object is 30cm from a spherical mirror along central axis. The absolute value of lateral magnification is 1/2 image produced is inverted what is the focal length and nature of mirror.

A) f = 10cm, convex B) f = -20cm, concave

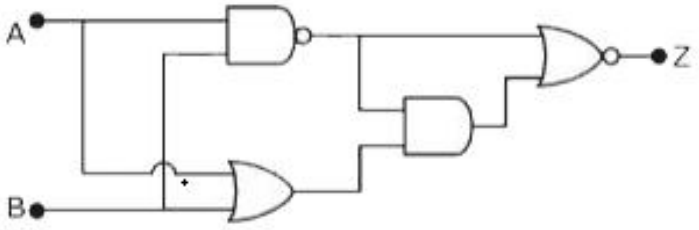
C) f = -10cm, concave D) f =20cm, convex

10. An electric dipole of two electric charges of 5μc magnitude each with separation 0.2μm is placed between the parallel plates of capacitor of capacitance 5μF and charges 2Q & -Q on its plates (Q =2μC). Separation between the plates 0.3mm. If the dipole is rotated by 30° from an axis that is parallel to electric field between plates of capacitor, dipole tends to realign in the direction due to a torque. The value of torque is

A) $5\times10^{-9}\,Nm$ B) $2\times10^{-9}\,Nm$ C) $1\times10^{-9}\,Nm$ D) $3\times10^{-9}\,Nm$

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11. In the following digital circuit, what will be the output at 'z', when the input [A,B] are [1,0] [0,0] [1,1] [0,1]



A) 0,1,0,1 B) 1,0,1,0 C) 0,0,1,0 D) 1,0,0,1

12. Match List-I with List-II

List-I		List-II	
A	Torque	I	ML^2T^0
B	Angular momentum	II	ML^2T^{-1}
C	Moment of inertia	III	M^0LT^0
D	Radius of gyration	IV	ML^2T^{-2}

A) A-IV, B-II, C-III, D-I B) A-I, B-III, C-IV, D-II

C) A-IV, B-II, C-I, D-III D) A-II, B-III, C-IV, D-I

13. A wave of frequency 500Hz has a velocity 360m/s. The distance between the nearest points which are 60° out of phase is.

A) 0.7cm B) 12cm C) 70cm D) 120cm

14. Assertion (A): If Refractive index of one medium is equal to refractive index of second medium, then light beam does not bend at all.

Reason (R):- The bending of light does not depend on refractive indices of media.

A) (A) is not correct but (R) is correct

B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

C) (A) is correct but (R) is not correct

D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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15. **Assertion (A):** Magnetic monopoles do not exist.

Reason (R): Net magnetic flux through any closed surface is zero.

A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

B) (A) is correct but (R) is not correct

C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

D) (A) is not correct but (R) is correct

16. Potential energy is defined for only

A) conservative force B) Non conservative force

C) pseudo force D) All forces

17.

List-I		List-II	
A	Cyclic	I	ΔQ(supplied heat)=0
B	Adiabatic	II	ΔU(change in Internal energy)$\neq 0$
C	Polytropic	III	$\Delta U=0$
D	Isobaric	IV	Pressure depends on volume

A) A-III, B-I, C-IV, D-II B) A-IV, B-I, C-III, D-II

C) A-III, B-I, C-II, D-IV D) A-I, B-IV, C-III, D-II

18. An aeroplane is flying in a horizontal direction with a velocity of 600 km/hr at a height of 1960m. when it is vertically above a point 'A' on the ground, a body is dropped from it. Body strikes the ground in time 't' at a point 'B'. Calculate time 't' & Displacement 'AB' [use g = 9.8m/s^2 neglect air resistance)

A) t=10sec AB = 6.66km B) t=20sec AB = 3.33km

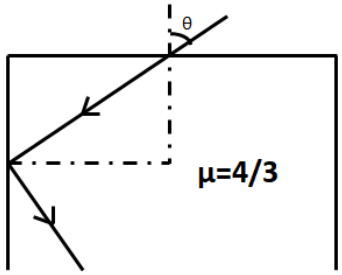
C) t = 10sec AB = 3.33km D) t=20sec AB = 6.66km

19. The motion of a particle of mass 500 gram is described by $y=5t+8t^2$. (t-time) The Force acting on particle is

A) 4N B) 8N C) 6N D) 5N

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20. A ray of light entering from air into a denser medium of refractive index 4/3, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle 'θ' should be equal to



A) $\sin^{-1}\frac{\sqrt{5}}{4}$ B) $\sin^{-1}\frac{\sqrt{7}}{3}$ C) $\text{Tan}^{-1}\frac{\sqrt{5}}{3}$ D) $\text{Tan}^{-1}\frac{\sqrt{7}}{3}$

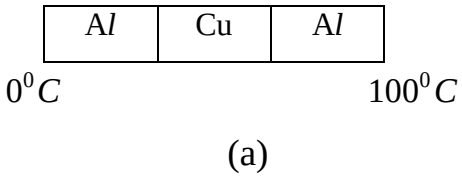
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A dielectric slab is inserted between the plates of capacitor almost fills the space between the plates. Dielectric constant of slab is '3'. If the induced charge on one face of slab is 8μC then parallel plate capacitor has charge of_____ μC.

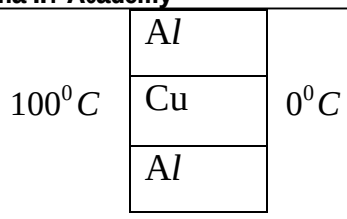
22. An inductor of reactance 100Ω, capacitor of reactance 20 Ω and a resistance 60 Ω are connected in series with an AC source of 220V, 50Hz. Power factor of circuit is 'x' find 10x.

23. The three rods shown in figure have identical dimensions. Heat flows from the hot end at a rate of 40W in the arrangement 'a'. Find the rate of heat flow in watt when the rods are joined as in arrangement 'b' [K_{Al} =200 W/m°c and K_{cu} = 400 W/m^{0}c]



(a)

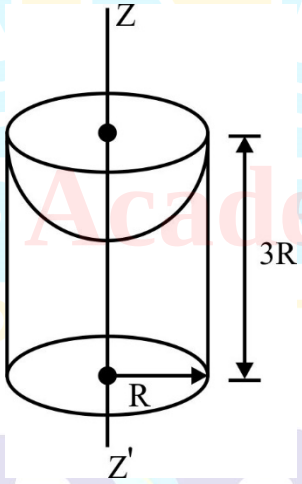
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(b)

24. If the electric field is given by $3\hat{i}+2\hat{j}+6\hat{k}$ N/C. The electric flux through a surface Area $2m^2$ lying in XY plane is _____ Nm^2C^{-1}

25. A solid cylinder is made of radius 'R' and height '3R' having mass density 'ρ'. Now one half sphere of radius 'R' is removed from one end. The moment of inertia of remaining portion about axis ZZ' can be calculated as $\frac{y}{30}\pi R^5\rho$. Find '**y**'.



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. **Statement (I):** only α- Amino acids are obtained on hydrolysis of proteins. These all naturally occurring α-Amino acids are optically inactive.

Statement(II): Natural proteins are denaturated by acids which convert the globular proteins from water soluble form to their water insoluble form.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) both statement I and statement II are correct.

B) statement I is incorrect but statement II is correct

C) both statement I and statement II are incorrect

D) statement I is correct but statement II is incorrect

27. Mixture of $1.0g$ each of phenol, aniline and bromo benzene is dissolved in ethyl acetate and placed in a separating funnel. Excess $NaOH$ was added in the same funnel. The funnel was shaken vigorously and allowed to settle. The Ethylacetate layer in the funnel contains:

A) Phenol and Aniline B) Bromo Benzene and phenol

C) Bromo Benzene and Aniline D) only phenol

28. The hydration enthalpy of Mg^{+2} and SO_4^{-2} are $-a$ and $-b\ kJ/mol$ respectively. If the heat of solution of $MgSO_4$ is $d\ kJ/mole$, then the lattice enthalpy of $MgSO_4$ is

A) $d-a+b$ B) $d+a+b$ C) $(a+b)-d$ D) $d-(a+b)$

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29. $A_{(g)} \rightarrow 2B_{(g)} + C_{(g)}$ is a first order reaction

Time	T	∞
P_{system}	P_t	P_∞

The reaction was started with reactant 'A' only, which of the following expression is correct for rate constant K ?

A) $K = \frac{1}{t}\ln\frac{3P_\infty}{2(P_\infty - P_t)}$ B) $K = \frac{1}{t}\ln\frac{3P_\infty}{P_\infty - P_t}$

C) $K = \frac{1}{t}\ln\left(\frac{2P_\infty}{2P_\infty - 3P_t}\right)$ D) $K = \frac{1}{t}\ln\frac{2P_\infty}{3(P_\infty - P_t)}$

30. A compound has molecular formula $C_6H_{12}O$. It does not reduce Tollen's (or) Fehling's Reagent. But it gives positive test with $2,4-DNP$ and on Clemenson Reduction gives 2-methyl pentane. The structural formula of the compound is

A) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\underset{CH_3}{\underset{|}{CH}}-CH_3$

B) $CH_3-CH_2-\underset{CH_3}{\underset{|}{CH}}-\overset{O}{\overset{\|}{C}}-CH_3$

C) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-H$

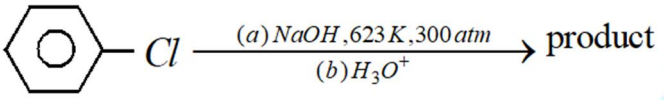
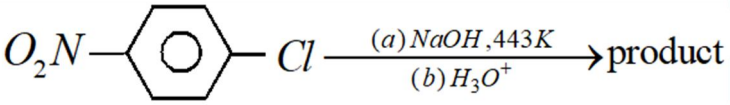
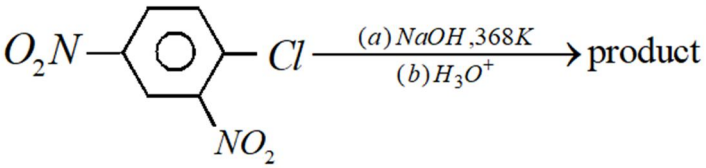
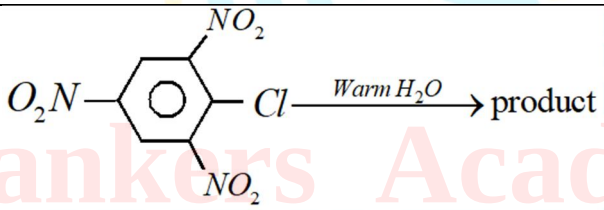
D) $CH_3-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$

31. Choose the correct trend in the atomic radii (r) of the elements and ions :

A) $r_{Na^+} > r_{Na}$ B) $r_{Al} < r_{Ga}$ C) $r_{Cl^-} > r_{Cl}$ D) $R_{K^+} < r_{Ca^{+2}}$

For More Join: @IITJEE_Advanc

32. Match List - I with List – II

	List – I Conversion		List – II Products pK_a - values
A		I	0.71
B		II	4.0
C		III	7.15
D		IV	9.98

Choose the correct answer from the options given below:

A) A-II, B-III, C-I, D-IV

B) A-III, B-IV, C-II, D-I

C) A-IV, B-III, C-II, D-I

D) A-IV, B-III, C-I, D-II

33. Choose the correct reaction for which the enthalpy of reaction is equal to the standard enthalpy of formation of product.

A) $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$

B) $2C_{(graphite)} + 3H_{2(g)} \rightarrow C_2H_{6(g)}$

C) $3O_{2(g)} \rightarrow 2O_{3(g)}$

D) $S_{8(S)_{Rhombic}} + O_{2(g)} \rightarrow SO_{2(g)}$

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34. Liquids A and B form an ideal solution in the entire composition range. At 350K. The vapour pressures of pure A and pure B are 7×10^3 Pa and 12×10^3 Pa Respectively. The composition of the vapour $(Y_A \text{ and } Y_B)$ in equilibrium with a solution containing 40 mole percent of A at this temperature is

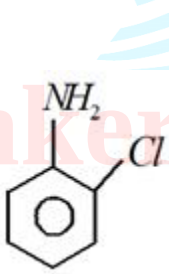
A) $Y_A = 0.76; Y_B = 0.24$ B) $Y_A = 0.28; Y_B = 0.72$

C) $Y_A = 0.4; Y_B = 0.6$ D) $Y_A = 0.37; Y_B = 0.63$

35. X is the number of coloured compounds due to charge transfer phenomenon among anhydrous $CuSO_4$, $KMnO_4$, K_2CrO_4, CrO_3, $MnSO_4$. The secondary valency of central metal ion in the complex $\left[Co(C_2O_4)_3\right]^{-3}$ is Y. The value of $Y - X$ is

A) 3 B) 4 C) 5 D) 6

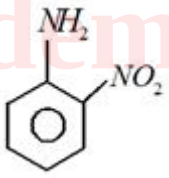
36. The descending order of basicity of following amines is



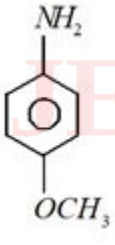
1)



2)



3)



4)

Choose the correct answer from the options given below:

A) $1 > 4 > 2 > 3$ B) $1 > 2 > 3 > 4$

C) $3 > 1 > 2 > 4$ D) $4 > 2 > 1 > 3$

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37. Match List - I with List – II

List – I		List – II		
Complex			Primary valency	Secondary valency
P)	$\left[Co\left(NH_3\right)_5 Cl\right]Cl_2$	I	2	4
Q)	$\left[Cu\left(en\right)_2\right]SO_4$	II	4	6
R)	$\left[Pt\left(NH_3\right)_5 Cl\right]Cl_3$	III	3	6
S)	$\left[Mg\left(EDTA\right)\right]^{-2}$	IV	2	6

Choose the correct answer from the options given below:

A) $P-III;Q-I;R-II;S-IV$

B) $P-III;Q-IV;R-II;S-I$

C) $P-IV;Q-I;R-III;S-II$

D) $P-IV;Q-II;R-III;S-I$

38. Match List - I with List – II

List - I		List – II	
P)	Ethanol + water	I	$\Delta H_{Mix}=0$
Q)	n-Hexane + n-Heptane	II	Minimum Boiling Azeotrope
R)	Nitric Acid + water	III	Negative deviation from ideal solution
S)	HCl + water	IV	Maximum Boiling Azeotrope

A) $P-IV;Q-II;R-I;S-III,IV$

B) $P-II;Q-IV;R-III,IV;S-I,IV$

C) $P-II;Q-I;R-III,IV;S-IV$

D) $P-I;Q-II;R-IV;S-III,IV$

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39. In CH_3^+, NO_3^- and I_2Cl_6 the hybridizations at the central atom are respectively:

A) Sp^2, Sp^2 and Sp^3d B) Sp^2, Sp^2 and Sp^3d^2

C) Sp^3, Sp^2 and Sp^3d^3 D) Sp^2, Sp^3 and Sp^3

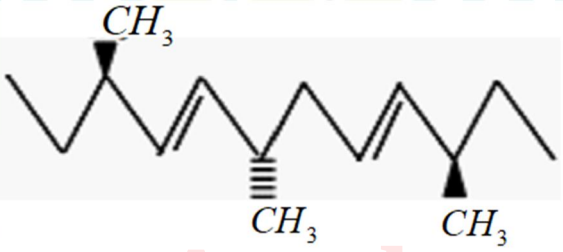
40. Benzaldehyde shows positive

(i) Tollen's test (ii) Fehiling's test

(iii) lodoform test (iv) DNP test

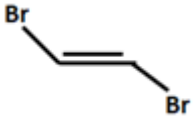
A) iii B) i and iv C) ii and iv D) ii and iii

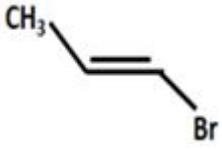
41. The number of different optically active products obtained during complete ozonolysis of following compound (not mirror images)



A) 2 B) 3 C) 4 D) 0

42. Given below are two statements:

Statements (I):  is more soluble in H_2O than 

Statements (II):  is more soluble in H_2O than 

In the light of the Above statements, choose the most Appropriate answer from the option given below.

A) statement I is correct but statement II is incorrect

B) statement I is incorrect and statement II is correct

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C) both statement I and statement II are incorrect

D) both statement I and statement II are correct

43. The extra stability of half-filled subshell is due to

(1) relatively smaller shielding of electrons by one another.

(2) smaller exchange energy

(3) the presence of electrons with the same spin in degenerate orbitals.

(4) unsymmetrical distribution of electrons.

Identity the correct statement:

A) 2,3 and 4 only B) 1 and 3 only

C) 2 and 3 only D) 3 and 4 only

44. The incorrect statements from the following are:

(1) Tl^{+3} is most stable due to inert pair effect

(2) Tl^{+3} is a powerful reducing agent.

(3) Al^{+3} and Tl^{+3} are highly stable

(4) Tl^{+} and Al^{+3} highly stable.

(5) Tl^{+1} is most stable due to strong oxidising Nature

choose the correct answer from the options given below:

A) 2,4 only B) 1, 3, and 4 only

C) 4 only D) 1, 2, 3 and 5 only

45. Given below are two statements:

1M aqueous solutions of each of $CuSO_4, AgNO_3, K_2SO_4, MgCl_2, NaOH$ are electrolysed, using inert electrodes

Statement(I): With increasing voltage, the sequence of deposition of metals on the cathode will be Na, Mg and K.

Statement(II): Cu and Ag metals not deposited at cathode instead of hydrogen gas will be evolved at the cathode.

In the light of the above statement, choose the most appropriate answer from the options given below:

A) both statement I and statement II are incorrect

B) statement I is correct but statement II is incorrect

For More Join: @IITJEE_Advanc

C) both statement I and statement II are correct

D) statement I is incorrect but statement II is correct.

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. The pH of an aqueous solution containing 1M benzoic acid $\left(Pk_a = 4.20\right)$ and 1M Sodium Benzoate is 4.5. The volume of benzoic acid aolution in 225 ml of this Buffer solution is_____ ml.

47. In KJeldahl's estimation of Nitrogen, the ammonia obtained from 0.8g of an organic compound was passed into 100ml of 0.2M H_2SO_4. The excess of acid required 80ml of 0.1M NaOH for neutralization. The percentage of Nitrogen in the compound is____

48. Propane reacts with oxygen to produce carbon dioxide and water following the equation given below

$$\mathbf{C_3H_{8(g)} + 5\ O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(l)}}$$

If 70kg of propane is mixed with 220kg of O_2. The volume of liquid water formed in liters is _____

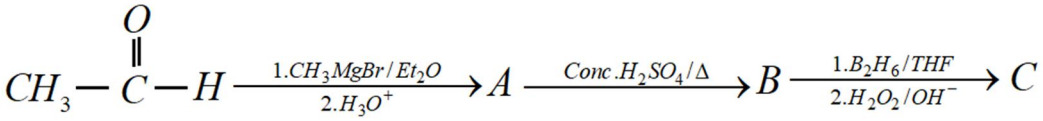
(Given : (a) Molar mass of C, H, O are 12, 1, 16g.mol^{-1} respectively,
(b) Density of water = 1g/ml)

49. Among the following given complexes, the number of paramagnetic species with same number of unpaired electrons is _____

$$\left[Zn\left(NH_3\right)_4\right]^{+2},\ \left[Ni\left(CO\right)_4\right],\ \left[Ni\left(CN\right)_4\right]^{-2},\ \left[Fe\left(CN\right)_6\right]^{-3},$$
$$\left[Cr\left(NH_3\right)_6\right]^{+3},\ \left[Fe\left(H_2O\right)_5 NO\right]SO_4,\ \left[CoF_6\right]^{-4},\left[Fe\left(H_2O\right)_6\right]^{+3}$$

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50. Identify the structure of the final product (C) in the following sequence of the reaction:



Find the total number of sp^3 hybridized atoms present in product "C" is.

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If from the point $P(4,4)$, perpendiculars to the straight lines $3x+4y+5=0$ and $y=mx+7$ meet at Q and R respectively and area of triangle PQR is maximum, then the value of $6m$ is equal to

A) 8 B) -6 C) 5 D) -5

52. Let $\vec{a},\vec{b},\vec{c}$ be three non coplanar vectors where $\vec{b_1}=\vec{b}-\left(\frac{\vec{b}.\vec{a}}{|\vec{a}|^2}\right)\vec{a}$ and

$\vec{c_1}=\vec{c}-\left(\frac{\vec{a}.\vec{c}}{|\vec{a}|^2}\right)\vec{a}-\left(\frac{\vec{b_1}.\vec{c}}{|\vec{b_1}|^2}\right)\vec{b_1}$ then:

A) $\vec{b_1}\times\vec{c_1}=\vec{0}$ B) $\vec{a}\times\vec{b_1}=\vec{0}$ C) $\vec{b_1}.\vec{c_1}=0$ D) $\vec{a}\times\vec{c_1}=\vec{0}$

53. Let $A=\{(\alpha,\beta)\in R\times R:|\alpha-1|\le 5,|\beta-6|\le 6\}$ and

$B=\{(\alpha,\beta)\in R\times R:16(\alpha-2)^2+9(\beta-6)^2\le 144\}$ then

A) $A\subset B$

B) $A\cup B=\{(x,y):-4\le x\le 4,-1\le y\le 11\}$

C) Neither $A\subset B$ nor $B\subset A$ D) $B\subset A$

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54. If $-3<\frac{x^2+ax-2}{x^2+x+1}<2$ holds for all real x, then exhaustive value of a is__

A) $[-2,1)$ B) $(-2,1)$ C) $R-[-2,2]$ D) $(-2,2)$

55. Box I contains 3 white and 2 red balls and Box II contains only 1 white ball. A fair coin is tossed. If head appears, then one ball is drawn at random from Box I and put into Box II. However, if tail appears, then two balls are drawn at random from Box I and put into Box II. Now one ball is drawn at random from Box II. It is given that the ball drawn from Box II is white, the probability that head appears on the coin, is

A) $\frac{17}{23}$ B) $\frac{11}{23}$ C) $\frac{15}{23}$ D) $\frac{12}{23}$

56. A random variable X takes values $-1,0,1$ and 2 with probabilities $\frac{1+3p}{4},\frac{1-p}{4},\frac{1+2p}{4}$ and $\frac{1-4p}{4}$ respectively where p varies over set of real numbers. Then the minimum and maximum values of mean of X are respectively

A) $-\frac{7}{4}$ and $\frac{1}{2}$ B) $-\frac{1}{16}$ and $\frac{5}{16}$ C) $-\frac{7}{4}$ and $\frac{5}{16}$ D) $-\frac{1}{16}$ and $\frac{5}{4}$

57. Let $f(x)=\min\left\{e^x,1+e^{-x},\frac{3}{2}\right\}$ for all real values of x. Area (in sq. units) bounded by $f(x)$ with x axis and the lines $x=l\text{n}\left(\frac{3}{2}\right),x=l\text{n}\,2$ is given by:

A) $l\text{n}\left(\frac{8}{3}\right)$ B) $l\text{n}\,8-l\text{n}\sqrt{3}$ C) $l\text{n}\left(\frac{8}{3\sqrt{3}}\right)$ D) $l\text{n}\,3\sqrt{3}-l\text{n}\,2$

58. If $f(x)$ is a polynomial of degree 6 which satisfies $\lim\limits_{x\to 0}\left(1+\frac{f(x)}{x^3}\right)^{\frac{1}{x}}=e^2$ and has local maximum at $x=1$ and local minimum at $x=0$ and $x=2$, then the value of $\left(\frac{5}{9}\right)^4 f\left(\frac{18}{5}\right)$

A) 24 B) 32 C) 12 D) 16

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59. Let $x \in \left(-\frac{\pi}{2}, \frac{7\pi}{2}\right)$ and $y \in R$, then the number of ordered pairs (x, y) which satisfy the inequality $2^{\sec^2 x} \cdot \left\{\sqrt{\frac{1}{2} - y^2 + y^4}\right\} \leq 1$ is___

A) 4 B) 8 C) 12 D) 16

60. Let $a_1, a_2, a_3, \ldots, a_{50}$ be 50 distinct numbers in A.P and $\sum_{r=1}^{50} (-1)^{r+1} (a_r)^2 = \left(\sqrt{\frac{5}{7}}\right)^{\frac{n}{2}} \left(a_1^2 - a_{50}^2\right)$ where $n \in N$, then the value of n is equal to

A) 4 B) 2 C) 8 D) 10

61. If the locus of $z \in C$ such that $z = \frac{3i - t}{2 + it}$ $(t \in R \, is \, a \, parameter)$ is a circle whose centre is (a, b) and radius is r, then find the value of $a + 4b + 8r$

A) 5 B) 6 C) 7 D)8

62. Rectangle ABCD has area 200 sq.units. An ellipse with area 200π sq. units passes through A and C and has foci at B and D. If the perimeter of rectangle is $16M$, then the value of M is

A) 10 B) 5 C) 20 D) 15

63. Let the curve $y = f(x)$ passes through the point $(4, -2)$ and satisfy the differential equation $y\left(x + y^3\right)dx - x\left(y^3 - x\right)dy = 0$. Then the value of $\int_{-1/2}^{1/2} [f(x)]dx$ is equal to..... where $[.]$ denote the greatest integer function

A) 0 B) $\frac{1}{2}$ C) $-\frac{1}{2}$ D) -1

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64. If the line of shortest distance between the lines

$\frac{x-1}{1}=\frac{y+1}{-1}=\frac{z+1}{1}$ and

$\frac{x+2}{-1}=\frac{y-1}{-1}=\frac{z-2}{1}$

passes through the points $(\alpha,3,\beta)$ then the value of $8(\alpha+\beta)$ is

A) 2 B) -2 C) 4 D) 8

65. There are 12 identical seats in the first row of a theatre of which 4 persons are to be seated. Let p denote the number of arrangements so that there should be at least 2 empty seats between any two persons and q denote the number of arrangements so that each person has exactly one neighbour. Then the value of $(p+q)$ is equal to

A) 1212 B) 1216 C) 1220 D) 1224

66. If a line with direction ratios $0,2,-1$ meet the lines $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$ and $\frac{x-1}{1}=\frac{y+2}{3}=\frac{z-2}{-2}$ at A and B respectively, then the length of the line segment AB is given by

A) $2\sqrt{5}$ B) $4\sqrt{2}$ C) $\sqrt{5}$ D) $3\sqrt{5}$

67. Find the number of integral solutions of the equation

$|x+1|-|x|+3|x-1|-2|x-2|=x+2$ where $-2025\le x\le 2025$

A) 2025 B) 4050 C) 2024 D) 4048

68. The locus of the centre of a variable circle touching two circles of radii r_1 and r_2 externally, which also touch each other externally, is a conic. If $\frac{r_1}{r_2}=3+2\sqrt{2}$ then the eccentricity of the conic is

A) $\frac{1}{3}$ B) $\sqrt{2}$ C) $\frac{1}{2}$ D) $2\sqrt{2}$

69. The sum of all distinct values of t for which the system of equations $(t-1)x+(3t+1)y+2tz=0,(t-1)x+(4t-2)y+(t+3)z=0,$ $2x+(3t+1)y+3(t-1)z=0$ has a non-zero solution is

A) 10 B) 5 C) 12 D) 3

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70. Let $a_1, a_2, a_3, \ldots$ be in geometric progression with $a_{101} = 25$ and $\sum_{i=1}^{201} a_i = 625$.

Then the value of $\sum_{i=1}^{201} \frac{1}{a_i}$ is equal to

A) 1 B) 2 C) 3 D) 4

SECTION – II
(Numerical Value Answer Type)

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71. If $f(x) = \lim_{n\to\infty} \frac{\left(1-\cos\left(1-\tan\left(\frac{\pi}{4}-x\right)\right)\right)(x+1)^n + \mu\sin\left(\left(n-\sqrt{n^2-8n}\right)x\right)}{x^2(x+1)^n + x}, x \neq 0$

is continuous at $x = 0$, then find the value of $f(0) + 2\mu$

72. If $\int \left(x^{2010} + x^{804} + x^{402}\right)\left(2x^{1608} + 5x^{402} + 10\right)^{\frac{1}{402}} dx$

$= \frac{1}{10\alpha}\left(2x^{2010} + 5x^{804} + 10x^{402}\right)^{\frac{\alpha}{402}}$ then find the value of α

73. If $\lim_{x\to 0} \frac{x + \sin x - x\cos x - \tan x}{x^n}$ exists and is non zero finite value, then the value of n is

74. let the circle C touch the line $x - y + 1 = 0$, have the centre on the positive $x-$ axis, and cut off a chord of length $\frac{4}{\sqrt{13}}$ along the line $2y = 3x + 1$. Let H be the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, whose one of the foci is the centre of C and the length of the transverse axis is the diameter of C, Then the value of $2a^2 + 3b^2$ is equal to

75. If $\sum_{r=0}^{100} (2r+1)\left({}^{100}C_r\right)^2 = k.{}^{200}C_{100}$, then the value of k is

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